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Effective Speed, Problems Based on Boats, Escalators

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Question 1

A boat takes 7 hours to go from A to C, through a midpoint B. But it takes 8 hours to go from A to B and then return from B to A. How long it would take to go from C to A?

A) 7 hours

B) 8 hours

C) 9 hours

D) None of these

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Question 2

Krishna walks on an escalator at a rate of 5 steps per second and reaches the other end in 10 seconds. While coming back, walking at the same speed he reaches the starting point in 40 seconds. What is the number of steps on the escalator?

A) 40

B) 60

C) 80

D) 120

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Question 3

A man rows 42 km with the stream in 7 hours and 60 km against the stream in 12 hours. What is the speed of the boat in still water?

A) 5.5 kmph

B) 9.5 kmph

C) 2.5 kmph

D) 7.5 kmph

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Question 4

A man can row a boat at a speed of 6 kmph in still water. If it takes him twice as long to row up as to row down the river, the rate of current in the stream would be

A) 4 kmph

B) 2 kmph

C) 3 kmph

D) 2.5 kmph

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Question 5

A man can row 15 km/h in still water and he finds that it takes him twice as much time to row up than as to row down the same distance in the river. The speed of the current (in km/h) is :

A) 6 km/h

B) 6.5 km/h

C) 4.5 km/h

D) 5 km/h

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Question 6

City A to City B is a downstream journey on a stream which flows at a speed of 5km/hr. Boats P and Q run a shuttle service between the two cities that are 300 kms apart. Boat P, which starts from City A has a still-water speed of 25km/hr, while boat Q, which starts from city B at the same time has a still-water speed of 15km/hr. When will the two boats meet for the first time? (this part is easy) When will they meet for the second time?

A) 7.5 hours and 15 hours

B) 7.5 hours and 18 hours

C) 8 hours and 18 hours

D) 7.5 hours and 20 hours

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Question 7

Rohan walked down a descending escalator and took 40 steps to reach the bottom. Sohan started simultaneously from the bottom, taking 2 steps to every 1 step taken by Rohan. Time taken by Rohan to reach the bottom from the top is the same as time taken by Sohan to reach the top from the bottom.How many steps more than Rohan did Sohan take before they crossed each other?

A) 50

B) 45

C) 20

D) None of these

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Question 8

Due to a technical glitch, the escalator in a shopping mall moves two steps upward and one step downward. Rahul, who is in a hurry to meet his friend, doesn't wait for the escalator to take him upstairs but instead walks on it. If he covers 60 steps in total and he takes 15 seconds for it, find the number of steps he would have covered had the escalator been functioning properly.

A) 45

B) 30

C) 40

D) Cannot be determined

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Question 9

Shyama and Vyom walk up an escalator (moving stairway). The escalator moves at a constant speed. Shyama takes three steps for every two of Vyom's steps. Shyama gets to the top of the escalator after having taken 25 steps, while Vyom (because his slower pace lets the escalator do a little more of the work) takes only 20 steps to reach the top. If the escalator were turned off, how many steps would they have to take to walk up?

A) 45

B) 30

C) 40

D) Cannot be determined

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Question 10

In a flight of 600 km, an aeroplane was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/h and the time of flight was increased by 30 minutes. Find the duration of the flight

A) 1 hour

B) 2 hours

C) 3 hours

D) 4 hours

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Question 11

When moving downstream, a boat can travel 2 km more than the distance that it could travel in still water in an hour. Find the distance that it could travel upstream in an hour.

A) 6 km less than the distance in downstream

B) 2 km less than the distance in still water

C) 4 km less than the distance in downstream

D) More than one of the above

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Question 12

A man can cross a downstream river by steamer in 40 minutes and same by boat in 1 hour. If the time of crossing the river in upstream direction by steamer is 50% more than downstream time by the steamer and the time required by boat to cross the same river by boat in upstream is 50% more than the time required in downstream by boat. what is the time taken for the man to cross the river downstream by steamer and then return to same place by boat half the way and by steamer the rest of the way?

A) 85 min

B) 115 min

C) 120 min

D) 125 min

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Question 13

Two boats start at the same instant to cross a river W metre wide. The faster boat reaches the other bank and returns back immediately. What are the distances travelled by them they meet, where the speeds of these boats are b1 & b2?

A) $2W/(b1+b2)$, $2W/(b1-b2)$

B) $2W/(b1+b2) \times b1$ and $2W/(b1+b2) \times b2$

C) $W/(b1+b2) \times b1$, $W/(b1+b2) \times b2$

D) data insufficient

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Question 14

Mariya was travelling in her boat when the wind blew her hat off and the hat started floating back downstream. The boat continued to upstream for twelve more minutes before Mariya realized that her hat had fallen off and turned back downstream. She caught up with that as soon as it reached starting point. Find the speed of river if Mariya's hat flew off exactly 31m from where she started :

A) 5 km/h

B) 6 km/h

C) 7.5 km/h

D) can't be determined

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Question 15

Ravi takes 40 seconds to walk up on an escalator which is moving upwards but he takes 60 seconds to walk up on an escalator which is moving downwards. How much time will he take to walk up if the escalator is not moving?

A) 50 s

B) 100 s

C) 48 s

D) 60 s

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THANK YOU

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